Project Preliminary Analysis (PPA) - Group 2

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Data

NYC311 is a non-emergency service that addresses issues for New York City residents, streamlining community concerns. To highlight its benefits, New York City shared a dataset of 311 Service Requests from 2010 onward, which is updated daily. Our focus is a subset from September 2016, containing 39 variables and 185,357 service requests. Among these, three numeric variables include the unique request key, latitude, and longitude of the request's location. The remaining character-type variables detail the complaint description, location, agency provider, and relevant dates. This dataset offers insights into how residents utilize the NYC311 service for various concerns.

Pre-Processing

There were several issues in the NYC311 dataset provided to us. The first problem we encountered was the number of redundant variables the dataset had. This could be either due to most of the observations in the column being missing or, for having very similar information to other columns/variables. Therefore, the first step we made was to create a threshold that would drop any column that had more than 60% of its content missing, resulting in the removal of 11 variables. A further 9 variables were also dropped due to having already similar info on other columns. Another issue in the dataset pertained to how the dates were set-up. All variables related to any date were in character type which would later cause problems in any visualization or hypothesis we would want to investigate. For this reason, we replaced the previous date variables into new ones with a POSIXct format. The data specifically in 'DueDate' was then updated to get the latest deadline date between DueDate and ResolutionUpdate date for each instance, which led to the dropping of the column. 'IssueDuration' (the duration between IssuedDate and ResolutionDate in seconds) and 'IssueBalance' (balance between DueDate and ResolutionDate in seconds) were also added to the dataset. Another column that transformed the dates into days of the week was added, since this information could be helpful for the graphical visualizations.

Unresolved cases also created missing values in the dataset, mainly in the 'ResolutionDate' column. This instance helped us create a new column that differentiated resolved complaints from unresolved ones ('IsResolved'). Three other variables were calculated that were focused on the duration of the requests. 'ResolutionExpDurationDays' (expected duration between 'IssuedDate' and 'DueDate' in days), 'ResolutionObvDurationDays' (duration between 'IssuedDate' and 'ResolutionDate' in days) and 'ResolutionDurationCategory' to categorize 'ComplaintTypes' by their resolution duration (1-3 days, 1-2 weeks, etc…). The 'Status' column was then removed due to becoming redundant.

The observations in the variable CategoryType were then transformed into 10 different options ('Noise Complaints', 'Health & Sanitation', 'Water & Plumbing Issues', 'Building & Housing Issues', 'Public Space Conditions', 'Consumer Issues', 'Social Issues', 'Urban Infrastructure', 'Other') for clearer and more comprehensive visualizations.

Rows with both 'Incident Address' and 'Longitude'/'Latitude' missing were also removed considering we had no way of knowing where the localization of the complaint occurred.

After pre-processing, the dataset that was worked on for the rest of the Project Preliminary Analysis (PPA) had 181372

observations and 22 variables. Of the 22 variables, 5 were numerical/integer type variables (‘Unique Key’,’Longitude’,’Latitude’, ‘Resolution Exp Duration Days’, ‘Resolution Obv Duration Days’), 3 of them had Date POSIXct data types (‘IssuedDate’, ‘ResolutionDate’, ‘DueDate’), 2 had a Difftime data structure (‘IssueDuration’, ‘IssueBalance’) and the remaining 12 had character/categorical data types (‘Agency’, ‘Agency Name’, ‘ComplaintType’, ‘Descriptor’, ‘Location Type’, ‘Incident ZIP’, ‘Incident Address’, ‘Resolution Description’, ‘Community Board’, ‘Borough’, ‘IsResolved’, ‘ResolutionDurationCategory’).

Exploratory Data Analysis (EDA)

In this section of the report, we created several plots to identify patterns in the data and detect irregularities.

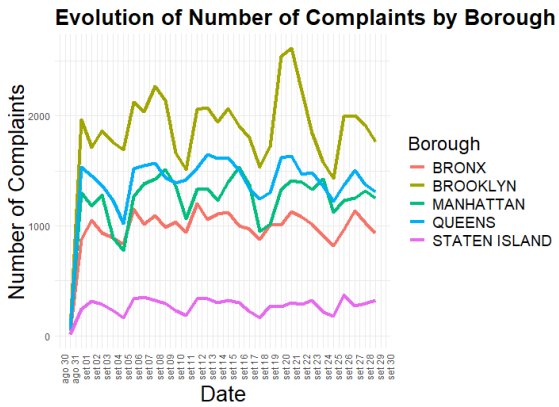


Figure 1. Evolution of the number of complaints per borough throughout the month. Shows that Brooklyn has the highest amount of complaints while Staten Island has the fewest.

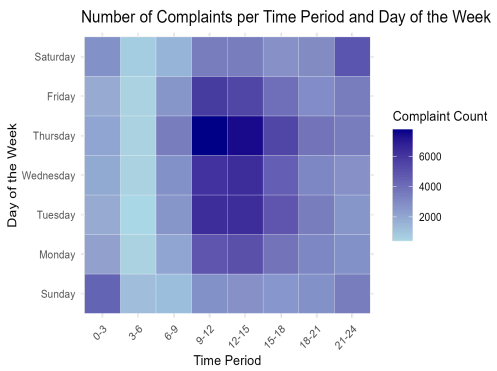


Figure 2. Number of complaints per week-day in different time periods of the day (hours). We can see that during the weekdays there is a higher amount of complaints in the morning/early afternoon and a significant increase in complaints on Saturday night.

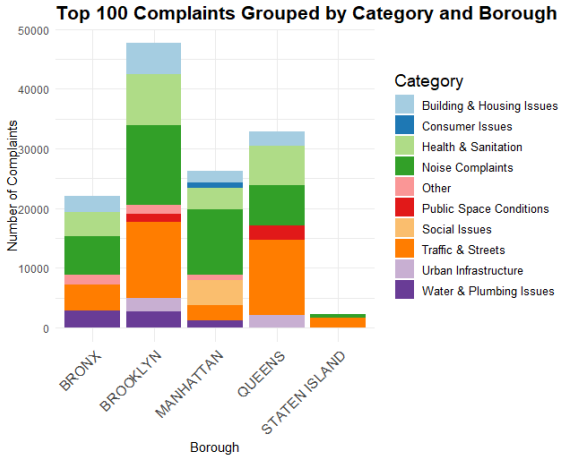


Figure 3. Categorized Reasons for the complaints per borough. In the Bronx, Brooklyn, and Manhattan, the majority of complaints are about noise, while in Queens and Staten Island, most complaints are related to traffic and streets. However, Brooklyn also has a significant number of complaints about traffic and streets.

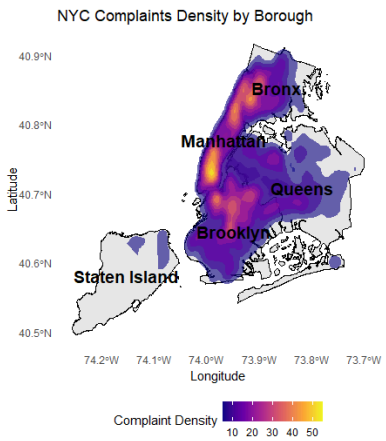


Figure 4. Density of complaints based on location within borough boundaries.

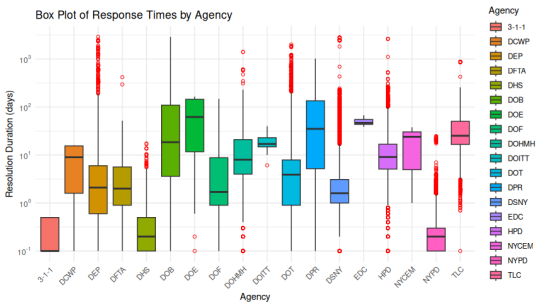


Figure 5. Average response time of the agencies. The Department of Buildings (DOB), the Department of Education (DOE) and the Department of Parks and Recreation (DPR) appear to be the agencies that take longer to resolve the complaints.

Hypothesis and Interesting Visualizations

Hypothesis A: *Weather conditions have a direct effect on the amount of requests made*

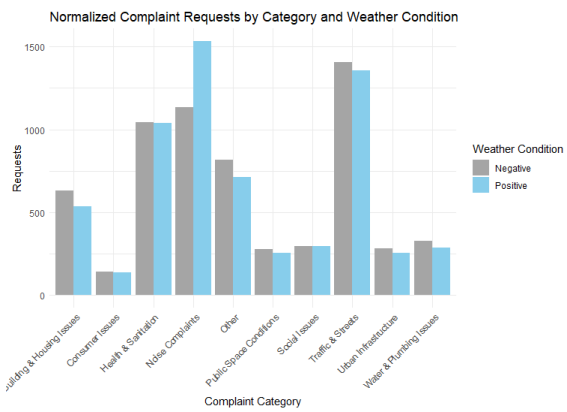


Figure 6. shows us how the weather can affect the type of requests that are made. The weather conditions of the month of September 2016, were obtained using "*timeanddate.com*". The number of requests for each Category Types on negative or positive weather conditions were normalized. The reason being the high discrepancy between the count of each could lead us to wrong conclusions. So, both of them needed to have the same weight for further analysis and comparison. Looking at the bar plot above we can see the 'Noise Complaints' were notably higher on days that were Sunny/Clear. In terms of the other categories, there is a slight tendency for more complaints on days with negative weather conditions. Mainly on the categories that could be affected by bad weather (Building & Housing Issues; Traffic & Streets).

Hypothesis B: *More complex requests take longer to resolve or have a higher amount of unresolved cases*

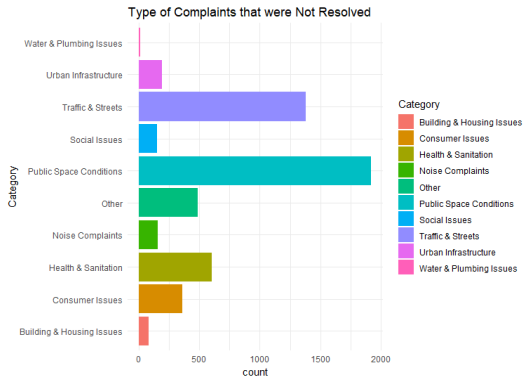


Figure 7. shows us that the number of unresolved complaints were higher in requests made in the categories 'Public Space Conditions' and 'Traffic & Streets' that would include sidewalk and road conditions. These could be because they are yet to be completed since agencies may have a later date set-up to resolve them.

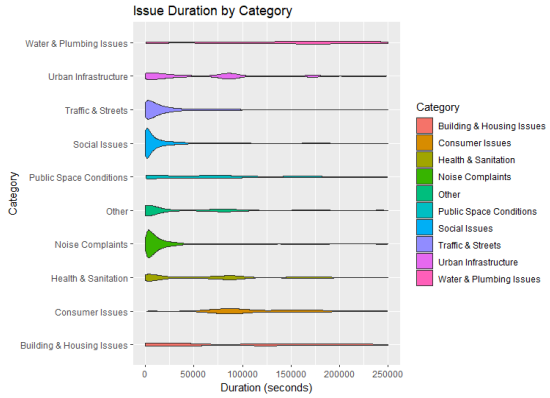


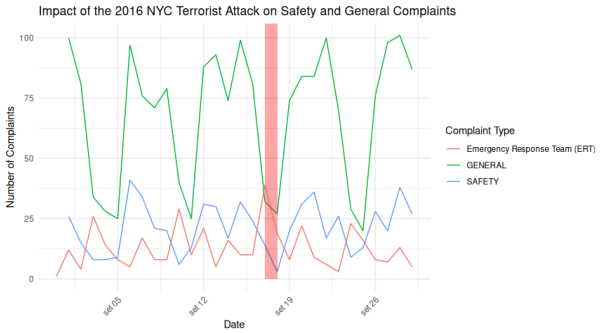
Figure 8. shows us the issue duration of each category. Looking at the plots we can see clearly that the categories that are more readily resolved consist of ones involved with noise complaints and Social issues that are usually easier since there is less labor needed to resolve them.

Hypothesis C: *The terrorist attacks of September 17th and 18th increased the safety concerns of the New Yorkers*

On 17–18 September 2016, four separate but connected terrorist-related bomb incidents occurred in the northeastern United States,

with two of them being bombs positioned at two sites in New York City and one managing to explode in the Manhattan district, wounding 31 people.

As a result, we tried to assess the effects of the attack on the inhabitants by checking whether there was any spike in safety/general complaints after the attack.



By looking at Figure 9, we can see the spike in Emergency Response Team (ERT) complaints on September 17th, which was the day of the first attack, and that points to the start of the contour regarding the span of the terrorist attacks. However, there was no evident change in the amount of safety or general complaints after the attack. It could be argued that the abrupt decrease in the complaints during that attack interval happened due to a probable lockdown mindset of the New Yorkers during the days of the attack, or that the instant increase in the complaints right after this period also could be correlated to the same inhabitants that closed in their homes returning to normality, but there's no strong evidence of an increase on the safety concerns after the attack so we can conclude that this hypothesis falls on the ground.

Hypothesis D : *There is a higher amount of Noise complaints on Sunday Night, mainly in bigger boroughs*

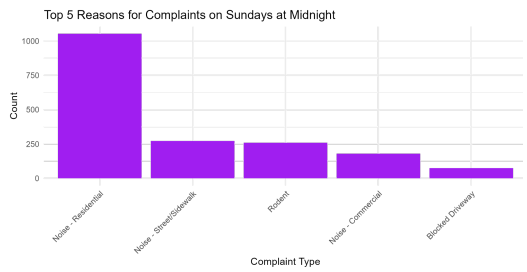


Figure 10. Shows that on Sunday Nights the main complaint is " Noise - Residential".

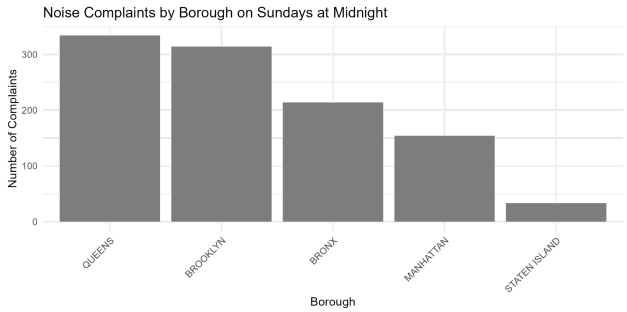


Figure 11. Queens, the second-largest borough by population, has the most residential noise complaints, while Brooklyn, the most populous borough in New York, has the second most. The Bronx is the fourth-largest borough in terms of population but ranks third in terms of noise complaints.

Future Work

For future work we believe that enhancing the information we can get from the visualizations to get clearer and interesting conclusions would be an interesting next step for this project. In hypothesis A, a clearer visualization could include hourly information on the weather condition with more descriptive options (Sunny, Clear, Heavy Rain, etc.), enhancing the visualization tremendously. In hypothesis B, a visualization of the normalization of either the resolution time taken or the volume of requests could make it easier to understand the distribution of the data points. Hypothesis C could use spreadsheets to create interactive dashboards, letting us filter complaint data by date, type, or response time, and analyze trends dynamically, and animations to track changes in complaint volumes and public satisfaction over time, especially around significant events such as an attack like this. Hypothesis D, could be benefited by making heat maps representing noise in the specific boroughs. Finally, different hypotheses could be explored, for further understanding of the problems that result in the complaint calls, with the goal of reaching solutions that would ultimately help the community reduce the need to request the NYC 311 service.